

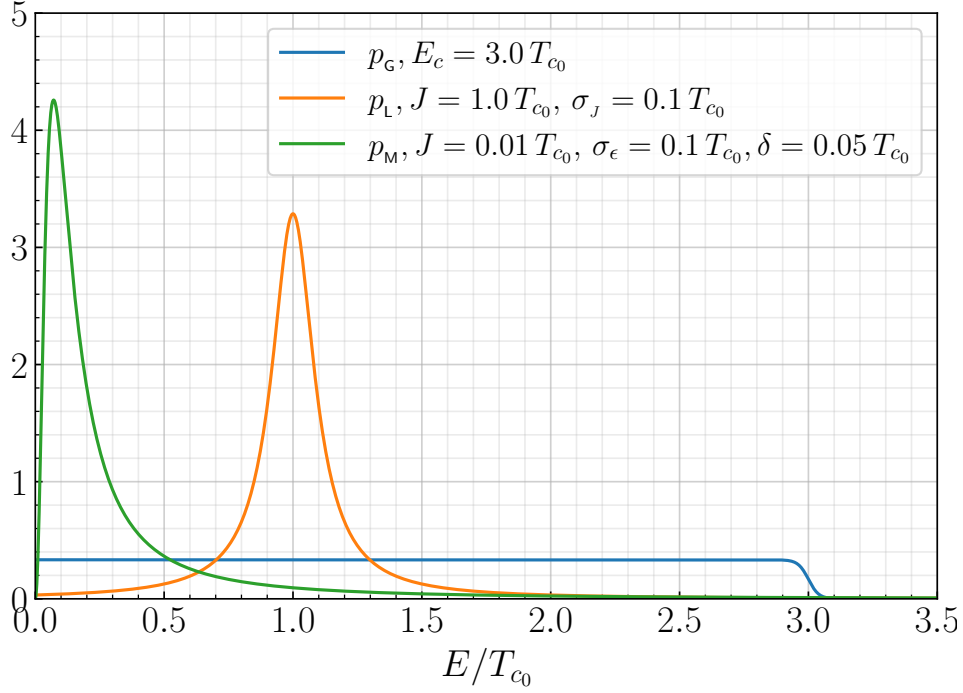


FIG. 5. Three models for the distribution of TLS tunnel splittings used to predict the effects of TLS impurities on superconducting properties. The distributions, $p_G(E)$, $p_L(E)$ and $p_M(E)$, are defined in Sec. IV C 2.

splittings, and (iii) a modified Lorentzian model introduced by Wipf and Neumaier for atomic tunneling in metals [6]. The glass model is defined by

$$p_G(E) = \frac{1}{E_c} \Theta(E_c - E), \quad (58)$$

where E_c is a high energy cutoff. Thus, for $T \ll E_c$ the relevant density of tunnel splittings is constant and explains many aspects of atomic tunneling in amorphous glasses [35, 36]. A key feature is there is substantial probability of TLS atoms with very low tunnel splitting.

For weaker disorder we consider a Lorentzian distribution of tunnel splittings based on the observation that small changes in the tunnel barrier or width of the double-well potential lead to significant changes in the tunnel splitting [5]. The Lorentzian distribution is defined by a most probable tunnel splitting, J , and a width of the distribution, σ_J ,

$$p_L(E) = \frac{1}{\pi} \frac{\sigma_J}{(E - J)^2 + \sigma_J^2}. \quad (59)$$

Thus, $\sigma_J \ll J$ corresponds to the limit of weak strain disorder, while $\sigma_J \gg J$ corresponds to substantial strain disorder impacting the distribution of tunnel splittings.

We also consider a modified version of the distribution introduced by Wipf and Neumaier for H and D tunneling in Nb. These authors assumed that the random strain field leads to a distribution of asymmetry energies for the double well potential, but does not effect the tunnel matrix element, J . The resulting tunnel splitting for any particular tunneling atom is $E = \sqrt{J^2 + \epsilon^2}$, where ϵ is the asymmetry energy, i.e. the difference between the energies of the minima of the double well potential. Wipf and Neumaier assumed a Lorentzian distribution for the asymmetry energies with width σ_ϵ , but assumed that the tunnel matrix element is the same for all TLS impurities. The resulting distribution is,

$$p_w(E) = \frac{2}{\pi} \frac{E}{\sqrt{E^2 - J^2}} \frac{\sigma_\epsilon}{E^2 - J^2 + \sigma_\epsilon^2} \Theta(E - J). \quad (60)$$

This model worked well in accounting for the anomalous heat capacity of low concentrations of O, H and D in Nb [6]. However, it has a square root singularity at a low-energy cut-off set by the tunnel matrix element J . We modify their distribution to account for variations in the tunnel matrix element due to strain disorder by introducing an imaginary part to remove the singularity and broaden the low-energy region associated with a distribution of tunnel matrix elements, i.e.

$$p_M(E) = N_w \text{Im} \frac{E}{\sqrt{J^2 - (E + i\delta)^2}} \frac{\sigma_\epsilon}{E^2 - J^2 + \sigma_\epsilon^2}, \quad (61)$$

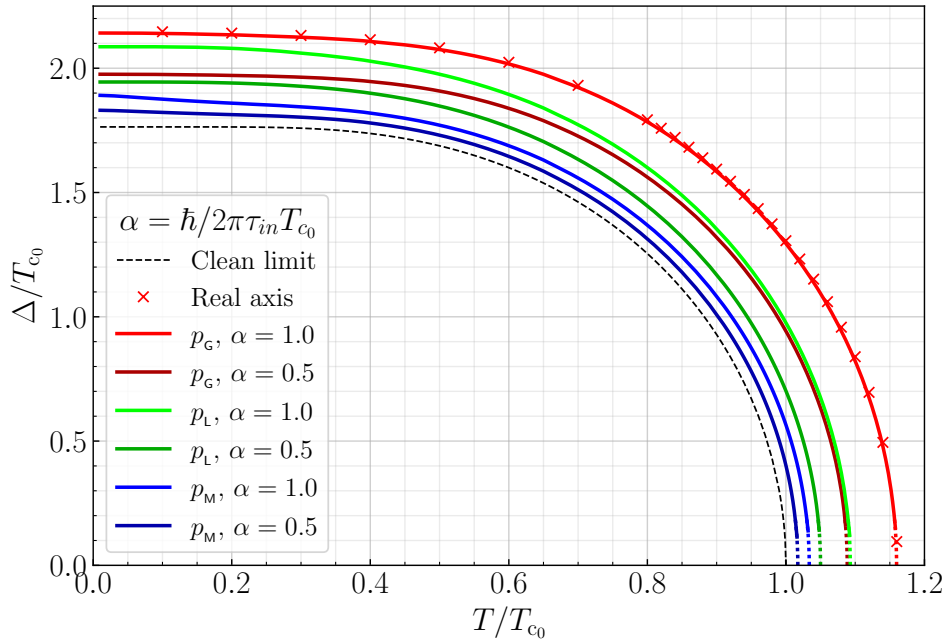


FIG. 6. Enhancement of T_c and the order parameter Δ versus temperature for the glass (p_G), Lorentzian (p_L) and modified Wipf-Neumaier (p_M) distributions of tunnel splittings, for inelastic scattering rates $\alpha \in \{0.5, 1.0\}$. The $\times$ symbols are solutions for Δ obtained from the gap equation defined on the real energy axis, for the glass distribution and $\alpha = 1.0$.

where N_w is the prefactor that ensures that $p_m(E)$ is properly normalized which is computed numerically for any set of parameters, J , σ_ϵ and δ .

The three distributions are shown in Fig. 5 with the cutoff in the glass distribution, $p_G(E)$, set at $E_c = 3T_{c0}$ corresponding to strong disorder with a broad distribution of tunnel splittings. For the Lorentzian distribution, $p_L(E)$, we set $J = T_{c0}$ and $\sigma = 0.1T_{c0}$, corresponding to relatively large tunnel splittings and weak disorder on the distribution of tunnel splittings. And for the modified Wipf-Neumaier distribution, $p_M(E)$, we set the peak in the distribution at relatively low energy, $J = 0.1T_{c0}$, broadening $\delta = 0.01T_{c0}$ and the spread in asymmetry energies of $\sigma = 0.1T_{c0}$. These parameters are close to those for H tunneling in Nb. Note that the Lorentzian and modified Wipf-Neumaier distributions have tails that decay as $1/E^2$.

V. EFFECTS OF TLS IMPURITIES ON SUPERCONDUCTIVITY

The thermodynamic properties of a superconductor with a concentration of TLS impurities depends on the order parameter obtained from a self-consistent solution of the gap equation Eq. (48) and the renormalization factors, Eqs. (49), (50) and (51), obtained from the inelastic contribution to the TLS-impurity self energy, (42). The effects of inelastic scattering by TLS impurities are determined by the distribution of tunnel splittings and the overall strength of the interaction between the TLS impurities and conduction electrons expressed in terms of the scattering rate $1/\tau_{in} = 2\pi n_s N(0) \langle |m_{\mathbf{k},\mathbf{k}'}|^2 \rangle_\Omega$, where n_s is mean density of TLS impurities, $N(0)$ is the normal-state quasiparticle density of states at the Fermi level and $\langle |m_{\mathbf{k},\mathbf{k}'}|^2 \rangle_\Omega$ is Fermi-surface average over the inelastic contribution to the quasiparticle-TLS scattering probability. It is useful to express this scattering rate in units of the rate set by the timescale, $\tau_0 = \hbar/2\pi T_{c0}$, for Cooper pair formation in the clean limit, i.e. $\alpha = \tau_0/\tau_{in} = \hbar/2\pi\tau_{in}T_{c0}$. Thus, $\alpha \ll 1$ corresponds to weak TLS impurity scattering, $\alpha \gg 1$ is the corresponding “dirty limit”, and $\alpha = 1$ is the intermediate disorder limit.

1. Equilibrium Order Parameter and T_c enhancement

Figure 6 shows results for the order parameter, $\Delta(T)$, obtained from self-consistent solutions of Eqs. (48), (49), (50) and (51), for the three models for the TLS distribution of tunnel splittings shown in Fig. 5, and for interaction strengths $\alpha = \hbar/2\pi\tau_{in}T_{c0}$ spanning weak to intermediate disorder.

In all three models inelastic scattering by TLS impurities leads to enhancement of superconductivity, both an increase in T_c as well as the order parameter (“gap”) at all temperatures. The gap enhancement is similar to that in superconductors with retardation from the electron-phonon interaction, c.f. Ref. [37] and references

therein. Indeed the largest enhancement of T_c and $\Delta(T)$ is exhibited by glass distribution which has the broadest distribution of tunnel splittings, and thus the largest relative ground state occupancy, N_{ge} . By contrast the modified Wipf-Neumaier distribution is peaked at $E = T_{c0}$, and thus has a significant population of TLS impurities in excited states. Distributions dominated by TLS level splittings with $E \geq T_{c0}$ give rise to larger gap enhancement.

For equilibrium TLS populations these impurities are predominantly in their ground state, and effectively static impurities. Thus, only the high energy tail of the distribution contributes to the gap enhancement.

The order parameter can also be calculated by analytically continuing Eq. (48) and the self energies to real energies. For TLS impurities in equilibrium with the electrons and lattice, the two formulations are equivalent and must give the same result for $\Delta(T)$. The comparison of the solutions based on the Matsubara gap equation and the real-frequency formulation are shown for the glass distribution and $\alpha = 1.0$; the results shown as the cross marks are in exact agreement with the solution obtained from the Matsubara gap equation.

2. Relation to earlier research on TLS impurities in superconductors

Early on several authors showed that T_c could be enhanced by impurities with discrete level splittings [22, 24, 38, 39]. Fulde, Hirst and Luther considered superconductors containing paramagnetic rare earth impurities with crystal-field split energy levels [38]. The latter leads to enhancement of T_c similar to that arising from optical phonons, and as we discussed inelastic scattering by TLS impurities. The enhancement competes with pair-breaking from spin-exchange scattering of conducting electrons by the paramagnetic impurity. Brandt also considered inelastic scattering by impurities with a level splitting, and obtained a result for the enhancement of T_c to first order in the impurity concentration, as a function of E/T_{c0} .

Vujičić et al. [39] considered TLS impurities embedded in the lattice of high- T_c superconductors, with each TLS impurity localized on the sites of an impurity lattice. Each TLS impurity was assumed to have the same tunnel splitting. In this model the localized TLS impurities are highly correlated in space, with long-range interactions. The authors then related inelastic electron-TLS scattering to an Eliashberg function, $\alpha^2(\omega)F(\omega)_{tls}$, and assumed the electron-TLS coupling to be the same magnitude as that for the coupling of band electrons to phonons of the pure superconductor. This is essentially an Einstein phonon spectrum except for dispersion of the TLS energy splittings spread over a narrow bandwidth, $\omega_{tls} \ll \omega_D$, compared to the phonon bandwidth ω_D . In the limit $T_c \ll \omega_{tls} \ll \omega_D$ the authors obtained the result

$$T_c = 1.14 \omega_{tls} \exp\{-1/\lambda_{\text{eff}}\} \text{ with } \lambda_{\text{eff}} = \lambda_{tls} + \frac{\lambda_{ph} - \mu^*}{1 - (\lambda_{ph} - \mu^*) \ln(\omega_D/\omega_{tls})}, \quad (62)$$

where $\lambda_a = \int d\omega \alpha^2 F(\omega)_a / \omega$ is the coupling constant for TLS or phonon mediated pairing, i.e. $a \in \{tls, ph\}$, and μ^* is the Coulomb pseudo-potential. The authors argue that for $\omega_{tls} \ll \omega_D$ and for nearly complete occupation of TLS impurities in the ground level, $N_g \gg N_e$, that $\lambda_{tls} \gg \lambda_{ph}$. This leads to extremely large enhancement of the transition temperature, $T_c/T_{c0} \simeq 27$, relative to that predicted by the electron-phonon interaction. The model and results differ dramatically from our predictions even for TLS impurities described by a sharp distribution of tunnel splittings. Given the model of a lattice of identical TLS impurities, and assumptions for the strength of the electron-TLS coupling, the overlap of this work with our theory is only the qualitative prediction that TLS impurities lead to an enhancement of T_c .

The formulation of the interaction of embedded TLS impurities in a superconductor by Maekawa et al. [24], based on tunneling impurities described by an SU(2) isospin, is closest to our formalism. However, these authors were interested in the Kondo limit in which the tunnel splitting $E \rightarrow 0^+$. In this limit the inelastic processes we consider are absent. Their results based on higher order scattering processes lead to weak pairbreaking and the suppression of T_c in the limit $E \ll T_{c0}$.

A main result of our microscopic theory for the interaction of a random distribution of TLS impurities on superconductivity, for a broad range to distributions of tunnel splittings, is that T_c and gap enhancement are determined by the high-energy part of the spectrum of TLS impurities, i.e. TLS impurities with $E \gtrsim T_{c0}$.

3. TLS induced superconductivity

The enhancement of the superconductivity by the random distribution of TLS impurities suggests that the kernel of the off-diagonal self energy, Σ_Δ , resulting from inelastic scattering by TLS impurities *generates* a retarded pairing interaction and corresponding TLS generated gap equation. [40] The latter is obtained from Eq.(49) with $\Delta = 0$,

$$\tilde{\Delta}(\varepsilon_n) = \frac{1}{2\pi\tau_{in}} \int dE p(E) N_{ge}(E) \pi T \sum_{\varepsilon_{n'}} \mathfrak{D}(\varepsilon_n - \varepsilon_{n'}; E) \frac{\tilde{\Delta}(\varepsilon_{n'})}{\sqrt{(\varepsilon_{n'} + i\Sigma_\epsilon(\varepsilon_{n'}))^2 + \tilde{\Delta}(\varepsilon_{n'})^2}}, \quad (63)$$